UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE

SEMESTER II - 2023/2024

MT 105 / MAT 1023 - REAL ANALYSIS I

END SEMESTER EXAMINATION

Time allowed: Three Hours **Answer ALL questions**

Registration number : ..... (I)

Instructions : This Examination consists of two parts. Clearly answer each part as instructed. Calculators are not allowed.

Part A — Multiple Choice Questions (Total: 60 marks)

This section consists of 10 multiple-choice questions, each carrying 6 marks. For each question, select the correct answer from the four options (A–D) and **clearly shade or circle the letter corresponding to your choice** in the question paper itself.

1. Let $E \subset \mathbb{R}$ be nonempty and bounded above. The least upper bound property asserts
A) $\sup E$ exists in $\mathbb{Q}$
B) $\sup E$ may not exist
C) $\sup E \in \mathbb{R}$ exists
D) $\sup E = \max E$

2. If $E \subset \mathbb{Z}$ is nonempty and bounded above, then
A) $\sup E$ may be non-integer
B) $\sup E \in E$
C) $\sup E \notin E$
D) $\sup E$ need not exist

3. The *Archimedean Principle* states that, for $a > 0$ and any $b \in \mathbb{R}$,
A) $\exists\ n \in \mathbb{Z}^+$ s.t. $na < b$
B) $\exists\ n \in \mathbb{Z}^+$ s.t. $b < na$
C) $\forall\ n \in \mathbb{Q}$ s.t. $na < b$
D) $\exists\ n \in \mathbb{Q}$ with $na = b$

4. Density of $\mathbb{Q}$ in $\mathbb{R}$ means
A) Between any two real numbers there is a rational number
B) Every real number is a rational number
C) Rational numbers are bounded
D) Only finitely many rationals in any interval

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5. If $\lim_{n\to\infty} x_n \to L$, then every subsequence $\{x_{n_k}\}$ satisfies

A) Diverges

B) Converges to a different limit

C) Converges to L

D) No Conclusion

6. Every bounded sequence in $\mathbb{R}$ has a

A) Unique limit

B) Non-unique limit

C) Convergent subsequence

D) Monotone subsequence

7. If $\lim_{n\to\infty} x_n \to 0$ and $\{y_n\}$ is bounded, then $\lim_{n\to\infty} x_n y_n \to$

A) ∞

B) 0

C) Does not exist

D) 1

8. The *Extreme Value Theorem* asserts

A) Continuous f on $[a, b]$ is bounded and attains max/min

B) Continuous f on (a, b) attains max/min

C) Differentiable f on $[a, b]$ is unbounded

D) Continuous f on $[a, b]$ has no max if not monotone

9. If the one-sided limits $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ both exist and are equal to L, then

A) $\lim_{x\to a} f(x)$ does not exist

B) $\lim_{x\to a} f(x) = L$

C) $f(a) = L$

D) f is discontinuous at a

10. If f is differentiable at a, then necessarily

A) f is continuous at a

B) f is monotone near a

C) $f'(a) > 0$

D) $f''(a)$ exists

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Part B — Structured Questions (Total: 340 marks)

There are three questions. Answer all three, providing clear, step-by-step justifications for each stage of your reasoning. Mark allocations for sub-parts are indicated in brackets.

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1. (a) Suppose that $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x+y) = f(x) + f(y)$ for each $x, y \in \mathbb{R}$.

   i. **(20 marks)** Prove that $f(nx) = nf(x)$ for all $x \in \mathbb{R}$ and $n \in \mathbb{Z}$.

   ii. **(20 marks)** Prove that $f(qx) = qf(x)$ for all $x \in \mathbb{R}$ and $q \in \mathbb{Q}$.

   iii. **(20 marks)** Prove that f is continuous at 0 if and only if f is continuous on $\mathbb{R}$.

   (b) **(5 marks)** State the *Monotone Convergence Theorem* (MCT).

   Define a sequence $\{x_n\}$ by

$$x_1 = \sqrt{5}, \quad x_2 = \sqrt{5+\sqrt{5}}, \quad x_{n+1} = \sqrt{5+x_n} \ \text{ for } \ n \geq 1.$$

   i. **(20 marks)** Use Mathematical Induction on n to show that $\{x_n\}$ is bounded above by 5.

   ii. **(20 marks)** Use Mathematical Induction on n to show that $\{x_n\}$ is increasing.

   iii. **(15 marks)** Using parts i and ii, find $\lim_{n\to\infty} x_n$.

2. (a) **(30 marks)** Use the precise $\epsilon - \delta$ definition to prove that $\lim_{x \to -6} \dfrac{x+4}{2-x} = -\dfrac{1}{4}$.

   (b) **(35 marks)** The *Dirichlet function* is defined on $\mathbb{R}$ by

$$f(x) := \begin{cases} 1 & : \ x \in \mathbb{Q}, \\ 0 & : \ x \notin \mathbb{Q}. \end{cases}$$

   Show that f is not continuous at any point of $\mathbb{R}$.

   (c) **(25 marks)**

   State the *Intermediate value theorem*.

   Prove that there is at least one $x \in \mathbb{R}$ which satisfies $e^x = 1 + 2\cos x$.

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3. (a) **(40 marks)** Clearly justify why the function

$$f(x) = \begin{cases} x^2 \sin\left(\dfrac{1}{x}\right) & x \neq 0, \\ 0 & x = 0. \end{cases}$$

is not continuously differentiable on any interval that contains the origin.
(*Hint :* Compute f' and use the 'Sequential criterion of Continuity').

(b) **(10 marks)** State the *Rolle's theorem* and *Mean value theorem*.

i. **(20 marks)** By using graphical representations, clearly explain the geometric interpretations of the two theorems.

ii. **(20 marks)** Use the *Rolle's theorem* to prove the *Mean value theorem*.

iii. **(20 marks)** Determine if the *Mean value theorem* can be applied to $f(x) = x^2 - x^{2/3}$ on the interval $[-1, 8]$. If so, find all possible values c that satisfy the conclusion of the *Mean value theorem*.

iv. **(20 marks)** Suppose that f is twice differentiable on (a, b) and that there are points $x_1 < x_2 < x_3$ in the interval (a, b) such that $f(x_1) > f(x_2)$ and $f(x_3) > f(x_2)$. Prove that there is a point $c \in (a, b)$ such that $f''(c) > 0$.

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